Document ID: VIB-OPS-FLD-010 Classification: INTERNAL CONFIDENTIAL Effective Date: 2024-04-01 Version: 3.2

Document Owner: National Head of Field Operations

Field Operations Capability Document: VI Business (India)

1. Executive Purpose and Document Scope

The Field Operations Capability Document details the logistical framework, deployment methodology, and physical maintenance strategy executed by the VI Business (India) Field Engineering teams. It proves our capacity to execute complex, multi-site enterprise deployments (such as nationwide SD-WAN rollouts) and maintain physical hardware SLAs in both dense urban hubs and remote rural territories.

This document serves as the internal baseline for answering RFP sections pertaining to "Site Installations," "Hardware Replacement (RMA)," and "Local Support Capabilities."

2. Definitions, Assumptions, and Applicability

Definitions:

- **RMA:** Return Merchandise Authorization (Hardware replacement process).
- **FTE:** Full-Time Equivalent (dedicated on-site resource).
- **CPE:** Customer Premises Equipment (e.g., edge routers, switches).
- **ZTP:** Zero-Touch Provisioning.

Assumptions:

- All deployment and break-fix SLAs defined herein assume the customer site is safely accessible by road and free from acute regional civil disturbances or natural disasters (force majeure).
- Target resolution times calculation begins only after the central Pune Tier-III NOC completes remote diagnostics and officially dispatches a field ticket.

Applicability:

- **Target Audience:** Bid Writers, Project Managers, and Enterprise Pre-sales Architects.
- **Scope:** Terrestrial fiber deployment, managed router installations, and IoT smart-metering physical deployments across India.

3. Site Survey Capability

Before complex physical fiber builds or multi-floor Wi-Fi deployments, VI Business mandates empirical feasibility checks.

- **Desktop Feasibility:** Using internal Geographic Information System (GIS) mapping overlays, the planning team validates optical fiber proximity within 4 hours of an Account Manager's query.

- **Physical Site Survey:** For sites requiring new civil works or Last-Mile fiber trenching, a physical survey engineer is dispatched.
    - **SLA:** Urban sites (within 48 hours), Rural sites (within 96 hours).
- **Deliverable:** A formal Site Feasibility Report detailing necessary civil permissions (e.g., municipal digging rights), required building duct pathways, and estimated lead times.

4. Installation and Dispatch Model

Our installation philosophy minimizes highly skilled engineering time on-site by shifting complexity to the centralized cloud.

4.1 The ZTP Installation Workflow (SD-WAN & Routing)

1. **Hardware Staging:** CPE is procured and centrally staged in the VI Mumbai Logistics Hub, where its serial number is mapped to the customer's cloud orchestrator tenant.
2. **Dispatch:** Unconfigured hardware is shipped directly to the remote site via secure national courier partners.
3. **Local Smart-Hands:** A localized VI Ecosystem Partner (certified technician) physically unboxes the device, mounts it in the rack, and connects the WAN circuits (Optical/LTE).
4. **Logical Provisioning:** The device dials the central Orchestrator, downloads its configuration, and comes online. No local command-line interface (CLI) configuration occurs.

5. Break-Fix Coverage and Response SLAs

When remote NOC troubleshooting fails to restore a down device, a Field Operations dispatch is triggered.

Fault Category	Dispatch Trigger	On-Site Response SLA (Urban / Tier 1)	On-Site Response SLA (Rural / Tier 3)
Physical Fiber Cut	OTDR distance confirmed via NOC.	4 Hours	8 Hours
Complete CPE Hard-Down	Power/Hardware failure confirmed.	Next Business Day (NBD)	Next Business Day (NBD)
Degraded Performance	Persistent packet loss unresolvable remotely.	48 Hours	72 Hours

Note: 4-Hour CPE replacement requires a dedicated, premium 'Forward Stocking Location' (FSL) contract.

6. Spare Parts (RMA) Model and Logistics

VI Business maintains a decentralized warehouse topology to ensure hardware replacement SLAs are viable across India's vast geography.

- **National Distribution Center (NDC):** Located in Bhiwandi (Maharashtra). Manages bulk OEM imports and central staging.
- **Regional Spares Depots (RSD):** Located in 16 major capital cities (e.g., Bengaluru, Delhi, Chennai, Kolkata). These depots stock "hot spares" for standard fortinet/Cisco SD-WAN routers, allowing Next Business Day (NBD) dispatch.
- **RMA Workflow:** Upon a declared hardware failure, the RSD ships the replacement device to the site. The local Field Engineer swaps the device, confirms ZTP logic restoration with the NOC, and packages the faulty unit for reverse logistics.

7. Regional Coverage Depth

VI Business maintains the densest enterprise field support footprint in the Indian telecom market.

- **Direct Field Engineers:** 1,200+ localized technical staff handling core optical splicing, cell-tower maintenance, and complex enterprise switching.
- **Authorized Ecosystem Partners:** A vetted network of 5,000+ certified "Smart Hands" technicians extending our reach into deep Tier-3 and Tier-4 townships.
- **Coverage Guarantee:** We ensure physical response capabilities across all 28 states and 8 Union Territories, supporting >95% of active Indian postal codes.

8. Field Governance and Escalation

To ensure partner quality matches direct VI standards, strict governance controls are enforced.

- **Digital Sign-Off:** All field interventions require the engineer to upload timestamped, geo-tagged photographs (e.g., neat rack cabling, lit interface LEDs) to the Field Service Management app before the ticket can be closed.
- **Escalation Matrix:** If an on-site engineer cannot resolve an anomaly within 2 hours of arrival, the ticket automatically escalates to the Regional Field Operations Manager and bridges the L3 Subject Matter Experts in Pune.

9. Constraints, Exclusions, and Dependencies

- **Customer Responsibilities:** The customer is strictly responsible for providing stable Uninterruptible Power Supply (UPS) and clear physical access to the server room. Field SLAs are paused if the engineer is denied entry by local customer security.
- **Civil Works Dependency:** Fiber restoration SLAs (e.g., 4 hours) are dependent on local police or municipal authorities granting immediate permission to excavate and splice.

10. Proposal-Use Guidance

- **Addressing Distributed Retail RFPs:** For banking or retail clients with hundreds of remote rural branches, forcefully cite Section 4 (ZTP Workflow) and Section 7 (Coverage Depth) to alleviate fears regarding difficult deployments.
- **SLA Defense:** Clearly delineate the difference between "NOC Response Time" (15 minutes) and "Physical On-Site Response Time" (Section 5) to avoid overcommitting in commercial contracts.

11. Governance and Ownership

- **Owner:** National Head of Field Operations
- **Reviewers:** Head of IT Service Delivery, CEBO
- **Update Cadence:** Annual review corresponding with logistics partner contract renewals.